

AI Assisted Coding

LAB ASSIGNMENT – 3.3

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Task 1: AI-Generated Logic for Reading Consumer Details

Scenario

An electricity billing system must collect accurate consumer data.

Task Description

Use an AI tool (GitHub Copilot / Gemini) to generate a Python program that:

- Reads:
 - Previous Units (PU)
 - Current Units (CU)
 - Type of Customer
- Calculates units consumed
- Implements logic directly in the main program (no functions)

Expected Output

- Correct input reading
- Units consumed calculation
- Screenshot showing AI-generated code
- Sample input and output

PROMPT:

```
"Write a Python script that reads previous units, current units, and customer type as input, calculates units consumed by subtracting the two values, "
"and prints the result, using direct logic in the main program without any functions."
```

Code:

```
previous_units = int(input("Enter previous units: "))
current_units = int(input("Enter current units: "))
customer_type = input("Enter customer type (residential/commercial): ")

units_consumed = current_units - previous_units

print("Customer Type:", customer_type)
print("Units Consumed:", units_consumed)
```

OUTPUT:

```
PS C:\Users\nagas> & "C:/Program Files/Python314/python.exe" "d:/college/3-2/AIAC/AIAC LAB/3.3.py"
Enter previous units: 120
Enter current units: 175
Enter customer type (residential/commercial): residential
Units consumed: 55
```

SUGGESTION:

The program correctly reads previous units, current units, and customer type from the user.

Units consumed are calculated accurately using direct logic in the main program without functions.

Sample input and output verify that the consumer details are read and processed correctly.

Task 2: Energy Charges Calculation Based on Units Consumed

Scenario

Energy charges depend on the number of units consumed and customer type.

Task Description

Review the AI-generated code from Task 1 and extend it to:

- Calculate Energy Charges (EC)
- Use conditional statements based on:
 - Domestic
 - Commercial
 - Industrial consumers
- Improve readability using AI prompts such as:
 - *"Simplify energy charge calculation logic"*
 - *"Optimize conditional statements"*

Expected Output

- Correct EC calculation
- Clear conditional logic
- Original and improved versions (optional)
- Sample execution results

PROMPT:

```
"Write a Python code to calculate Energy Charges based on units consumed and customer type (Domestic, Commercial, Industrial) using optimized conditional statements"
```

CODE:

```

14
15 previous_units = int(input("Enter previous units: "))
16 current_units = int(input("Enter current units: "))
17 customer_type = input("Enter customer type (Domestic/Commercial/Industrial): ")
18 units_consumed = current_units - previous_units
19 if customer_type == "Domestic":
20     if units_consumed <= 100:
21         charge = units_consumed * 1.5
22     elif units_consumed <= 300:
23         charge = (100 * 1.5) + (units_consumed - 100) * 2.5
24     else:
25         charge = (100 * 1.5) + (200 * 2.5) + (units_consumed - 300) * 4.0
26 elif customer_type == "Commercial":
27     if units_consumed <= 100:
28         charge = units_consumed * 2.0
29     elif units_consumed <= 300:
30         charge = (100 * 2.0) + (units_consumed - 100) * 3.5
31     else:
32         charge = (100 * 2.0) + (200 * 3.5) + (units_consumed - 300) * 5.0
33 elif customer_type == "Industrial":
34     if units_consumed <= 100:
35         charge = units_consumed * 2.5
36     elif units_consumed <= 300:
37         charge = (100 * 2.5) + (units_consumed - 100) * 4.0
38     else:
39         charge = (100 * 2.5) + (200 * 4.0) + (units_consumed - 300) * 6.0
40 print("Customer Type:", customer_type)
41 print("Units Consumed:", units_consumed)
42 print("Energy Charges: Rs.", charge)
43
44

```

OUTPUT:

```

PS C:\Users\nagas> & "C:/Program Files/Python314/python.exe" "d:/college/3-2/AIAC/AIAC LAB/3.3.py"
Enter previous units: 200
Enter current units: 260
Enter current units: 260
Enter customer type (Domestic/Commercial/Industrial): Domestic
Customer Type: Domestic
Units Consumed: 60
Energy Charges: Rs. 90.0

```

SUGGESTIONS:

The program correctly calculates Energy Charges based on units consumed and customer type.

Conditional statements are used clearly for Domestic, Commercial, and Industrial consumers.

The improved logic is easy to read and sample execution results confirm correct output.

Task 3: Modular Design Using AI Assistance (Using Functions)

Scenario

Billing logic must be reusable for multiple consumers.

Task Description

Use AI assistance to generate a Python program that:

- Uses user-defined functions to:
 - Calculate Energy Charges
 - Calculate Fixed Charges
- Returns calculated values
- Includes meaningful comments

Expected Output

- Function-based Python program
- Correct EC and FC values
- Screenshots of AI-assisted function generation
- Test cases with outputs

PROMPT:

```
'''Write a Python code using user-defined functions to calculate and return Energy Charges and Fixed Charges with multiple consumers'''
```

CODE:

```
def calculate_energy_charges(units_consumed, customer_type):
    if customer_type == "Domestic":
        if units_consumed <= 100:
            charge = units_consumed * 1.5
        elif units_consumed <= 300:
            charge = (100 * 1.5) + (units_consumed - 100) * 2.5
        else:
            charge = (100 * 1.5) + (200 * 2.5) + (units_consumed - 300) * 4.0
    elif customer_type == "Commercial":
        if units_consumed <= 100:
            charge = units_consumed * 2.0
        elif units_consumed <= 300:
            charge = (100 * 2.0) + (units_consumed - 100) * 3.5
        else:
            charge = (100 * 2.0) + (200 * 3.5) + (units_consumed - 300) * 5.0
    elif customer_type == "Industrial":
        if units_consumed <= 100:
            charge = units_consumed * 2.5
        elif units_consumed <= 300:
            charge = (100 * 2.5) + (units_consumed - 100) * 4.0
        else:
            charge = (100 * 2.5) + (200 * 4.0) + (units_consumed - 300) * 6.0
    return charge
```

```
def calculate_fixed_charges(customer_type):
    if customer_type == "Domestic":
        return 50
    elif customer_type == "Commercial":
        return 100
    elif customer_type == "Industrial":
        return 150
num_consumers = int(input("Enter number of consumers: "))
for _ in range(num_consumers):
    previous_units = int(input("Enter previous units: "))
    current_units = int(input("Enter current units: "))
    customer_type = input("Enter customer type (Domestic/Commercial/Industrial): ")
    units_consumed = current_units - previous_units
    energy_charges = calculate_energy_charges(units_consumed, customer_type)
    fixed_charges = calculate_fixed_charges(customer_type)
    total_charges = energy_charges + fixed_charges
    print("Customer Type:", customer_type)
    print("Units Consumed:", units_consumed)
    print("Energy Charges: Rs.", energy_charges)
    print("Fixed Charges: Rs.", fixed_charges)
    print("Total Charges: Rs.", total_charges)
    print("-----")
```

OUTPUT:

```
PS C:\Users\nagas> & "C:/Program Files/Python314/python.exe" "d:/college/3-2/AIAC/AIAC LAB/3.3.py"
Enter number of consumers: 2
Enter previous units: 200
Enter current units: 260
Enter customer type (Domestic/Commercial/Industrial): Commercial
Customer Type: Commercial
Units Consumed: 60
Energy Charges: Rs. 120.0
Fixed Charges: Rs. 100
Total Charges: Rs. 220.0
-----
Enter previous units: 120
Enter current units: 180
Enter customer type (Domestic/Commercial/Industrial): Industrial
Customer Type: Industrial
Units Consumed: 60
Energy Charges: Rs. 150.0
Fixed Charges: Rs. 150
Total Charges: Rs. 300.0
-----
```

SUGGESTIONS:

The program uses user-defined functions to calculate Energy Charges and Fixed Charges, making the logic reusable.

Each function returns the calculated value and includes meaningful comments for better understanding.

Test cases confirm that the functions produce correct EC and FC values for different consumers.

Task 4: Calculation of Additional Charges

Scenario

Electricity bills include multiple additional charges.

Task Description

Extend the program to calculate:

- FC – Fixed Charges
- CC – Customer Charges
- ED – Electricity Duty (percentage of EC)

Use AI prompts like:

- *“Add electricity duty calculation”*
- *“Improve billing accuracy”*

Expected Output

- Individual charge values printed
- Correct duty calculation
- Well-structured output
- Verified intermediate results

PROMPT:

```
"Write a Python code to calculate electricity bill charges including Energy Charges (EC), Fixed Charges (FC), Customer Charges (CC), and Electricity Duty "(ED as a percentage of EC), and display all charges clearly"
```

CODE:

```
previous_units = int(input("Enter previous units: "))
current_units = int(input("Enter current units: "))
customer_type = input("Enter customer type (Domestic/Commercial/Industrial): ")
units_consumed = current_units - previous_units
if customer_type == "Domestic":
    if units_consumed <= 100:
        energy_charges = units_consumed * 1.5
    elif units_consumed <= 300:
        energy_charges = (100 * 1.5) + (units_consumed - 100) * 2.5
    else:
        energy_charges = (100 * 1.5) + (200 * 2.5) + (units_consumed - 300) * 4.0
    fixed_charges = 50
    customer_charges = 20
    electricity_duty = energy_charges * 0.05
elif customer_type == "Commercial":
    if units_consumed <= 100:
        energy_charges = units_consumed * 2.0
    elif units_consumed <= 300:
        energy_charges = (100 * 2.0) + (units_consumed - 100) * 3.5
    else:
        energy_charges = (100 * 2.0) + (200 * 3.5) + (units_consumed - 300) * 5.0
    fixed_charges = 100
    customer_charges = 50
    electricity_duty = energy_charges * 0.10
elif customer_type == "Industrial":
    if units_consumed <= 100:
        energy_charges = units_consumed * 2.5
    elif units_consumed <= 300:
        energy_charges = (100 * 2.5) + (units_consumed - 100) * 4.0
    else:
        energy_charges = (100 * 2.5) + (200 * 4.0) + (units_consumed - 300) * 6.0
    fixed_charges = 150
    customer_charges = 100
    electricity_duty = energy_charges * 0.15
total_charges = energy_charges + fixed_charges + customer_charges + electricity_duty
```

```
print("Customer Type:", customer_type)
print("Units Consumed:", units_consumed)
print("Energy Charges: Rs.", energy_charges)
print("Fixed Charges: Rs.", fixed_charges)
print("Customer Charges: Rs.", customer_charges)
print("Electricity Duty: Rs.", electricity_duty)
print("Total Charges: Rs.", total_charges)
```

OUTPUT:

```
> & "C:/Program Files/Python314/python.exe" "d:/college/3-2/AIAC/AIAC LAB/3.3.py"
Enter previous units: 120
Enter current units: 780
Enter customer type (Domestic/Commercial/Industrial): Industrial
Customer Type: Industrial
Units Consumed: 660
Energy Charges: Rs. 3210.0
Fixed Charges: Rs. 150
Customer Charges: Rs. 100
Electricity Duty: Rs. 481.5
Total Charges: Rs. 3941.5
```

SUGGESTIONS:

The program correctly calculates Fixed Charges, Customer Charges, and Electricity Duty based on energy charges. All additional charges are calculated separately, making the billing logic clear and easy to follow. The output displays each charge clearly, allowing easy results.

Task 5: Final Bill Generation and Output Analysis

Scenario

The final electricity bill must present all values clearly.

Task Description

Develop the final Python application to:

- Calculate total bill:
- $\text{Total Bill} = \text{EC} + \text{FC} + \text{CC} + \text{ED}$
- Display:
 - Energy Charges (EC)
 - Fixed Charges (FC)
 - Customer Charges (CC)
 - Electricity Duty (ED)
 - Total Bill Amount
- Analyze the program based on:
 - Accuracy
 - Readability
 - Real-world applicability

Expected Output

- Complete electricity bill output
- Neatly formatted display
- Sample input/output
- Short analysis paragraph

PROMPT:

```
'''Write a Python code to calculate and display EC, FC, CC, ED, and the total electricity bill with sample input/output and a brief analysis of accuracy, readability, and real-world use'''
```

CODE:

```
# Function to calculate Energy Charges (EC)
def calculate_energy_charges(units, customer_type):
    if customer_type == "Domestic":
        if units <= 100:
            return units * 1.5
        elif units <= 300:
            return (100 * 1.5) + (units - 100) * 2.5
        else:
            return (100 * 1.5) + (200 * 2.5) + (units - 300) * 4.0
    elif customer_type == "Commercial":
        if units <= 100:
            return units * 2.0
        elif units <= 300:
            return (100 * 2.0) + (units - 100) * 3.5
        else:
            return (100 * 2.0) + (200 * 3.5) + (units - 300) * 5.0
    elif customer_type == "Industrial":
        if units <= 100:
            return units * 2.5
        elif units <= 300:
            return (100 * 2.5) + (units - 100) * 4.0
        else:
            return (100 * 2.5) + (200 * 4.0) + (units - 300) * 6.0

# Function to calculate Fixed Charges (FC)
def calculate_fixed_charges(customer_type):
    if customer_type == "Domestic":
        return 50
    elif customer_type == "Commercial":
        return 100
    elif customer_type == "Industrial":
        return 150
```

```
# Function to calculate Fixed Charges (FC)
def calculate_fixed_charges(customer_type):
    if customer_type == "Domestic":
        return 50
    elif customer_type == "Commercial":
        return 100
    elif customer_type == "Industrial":
        return 150

# Function to calculate Customer Charges (CC)
def calculate_customer_charges(customer_type):
    if customer_type == "Domestic":
        return 20
    elif customer_type == "Commercial":
        return 50
    elif customer_type == "Industrial":
        return 100

# Function to calculate Electricity Duty (ED)
def calculate_electricity_duty(energy_charges, customer_type):
    if customer_type == "Domestic":
        return energy_charges * 0.05
    elif customer_type == "Commercial":
        return energy_charges * 0.10
    elif customer_type == "Industrial":
        return energy_charges * 0.15

# ----- Main Program -----
n = int(input("Enter number of consumers: "))

for i in range(n):
    print("\n--- Consumer", i + 1, "---")
```

```
# ----- Main Program -----
n = int(input("Enter number of consumers: "))

for i in range(n):
    print("\n--- Consumer", i + 1, "---")

    previous_units = int(input("Enter previous units: "))
    current_units = int(input("Enter current units: "))
    customer_type = input("Enter customer type (Domestic/Commercial/Industrial): ")

    units_consumed = current_units - previous_units

    EC = calculate_energy_charges(units_consumed, customer_type)
    FC = calculate_fixed_charges(customer_type)
    CC = calculate_customer_charges(customer_type)
    ED = calculate_electricity_duty(EC, customer_type)

    total_bill = EC + FC + CC + ED

    # Output
    print("\n----- Electricity Bill -----")
    print("Customer Type      :", customer_type)
    print("Units Consumed      :", units_consumed)
    print("Energy Charges (EC)  : Rs.", EC)
    print("Fixed Charges (FC)   : Rs.", FC)
    print("Customer Charges     : Rs.", CC)
    print("Electricity Duty     : Rs.", ED)
    print("-----")
    print("Total Bill Amount    : Rs.", total_bill)
```

OUTPUT:

```
Enter number of consumers: 2

--- Consumer 1 ---
Enter previous units: 200
Enter current units: 280
Enter customer type (Domestic/Commercial/Industrial): Domestic

----- Electricity Bill -----
Customer Type      : Domestic
Units Consumed     : 80
Energy Charges (EC) : Rs. 120.0
Fixed Charges (FC)  : Rs. 50
Customer Charges    : Rs. 20
Electricity Duty    : Rs. 6.0
-----
Total Bill Amount   : Rs. 196.0

--- Consumer 2 ---
Enter previous units: 890
Enter current units: 1235
Enter customer type (Domestic/Commercial/Industrial): Industrial

----- Electricity Bill -----
Customer Type      : Industrial
Units Consumed     : 345
Energy Charges (EC) : Rs. 1320.0
Fixed Charges (FC)  : Rs. 150
Customer Charges    : Rs. 100
```

BRIEF ANALYSIS :

Accuracy:

The program correctly calculates energy charges and other charges to give the final bill.

Readability:

The use of separate functions makes the program easy to read and understand.

Real-World Applicability:

The program works like a real electricity bill system and can be used in practical situations.

SUGGESTION:

The program correctly calculates Energy Charges, Fixed Charges, Customer Charges, Electricity Duty, and the total bill.

Brief Analysis: The code is easy to read and understand because it uses clear logic and separate functions for each calculation.

The program follows a real electricity billing process, making it useful in practical situations.